expected to underperform the market. Achieving market neutrality from an investment perspective usually involves commitments of equal amounts of capital to over- and underperforming stocks, though in systematic trading market neutrality is achieved solely through exploitation of systematic risks of the stocks. The key idea here is to neutralize the portfolio against broad market moves, achieved by the offset of long positions' price sensitivity by short positions' price sensitivity (Jacobs and Levy, 2004).

Market-neutral strategies are underpinned by the capital asset pricing model (CAPM) developed by Sharpe (1964); Lintner (1965); Treynor (1961); and Mossin (1966). CAPM is given by:

$$\mathrm{E}[R_i] = R_f + \beta_i(\mathrm{E}[R_M] - R_f) \tag{13.18}$$

where R_i is the return on the capital asset, R_f is the risk-free rate, R_M is the return on the market portfolio and β_i is a measure of the systematic risk of the capital asset i relative to the market portfolio. CAPM is typically solved using OLS. The market portfolio is understood to be a portfolio consisting of market assets and the corresponding return is defined as the market return. Taking long-short positions with the same beta neutralizes systematic risk (having a beta of zero). Self-evidently such a construction is not risk-free though it will provide positive returns in recompense for accrued risk.

A common extension includes measures of investment managers' performance. Alpha is the intrinsic return on a capital asset and is used to measure returns in excess of the market. Taking the market regression of CAPM we have:

$$R_{i,t} - R_f = \alpha_i + \beta_i(R_{M,t} - R_f) + \varepsilon_t \tag{13.19}$$

Following Aldridge (2009), we shall proceed to describe a common market-neutral trading strategy.

Systematically trading the market-neutral pair of securities i and j involves determining the statistical significance means tests between the alphas and betas:

$$\hat{\beta} = \hat{\beta}_i - \hat{\beta}_j \tag{13.20}$$

$$\hat{\sigma}_{\Delta\beta} = \sqrt{\frac{\sigma^2_{\beta i}}{n_i} + \frac{\sigma^2_{\beta j}}{n_j}} \tag{13.21}$$